BRIDGE_C_35 order6

order6 REPORT (descriptive notes; verdicts as in RESULTS.md)

PREREG sha256 c19500e3…dfea (unchanged). Script hashes: scripts/SHA256SUMS. Deviations are listed in NOTES.md, and the recorded verdicts stand.

Category A (derived; proof sketches flagged)

- **Orthant split.** For generic α, the ν-sign orthant weight decomposes as orth_ν = ¼(1+s1)(1+s2) = ½·ODD + ½·EVEN, with ODD = ½(s1+s2) and EVEN = ½(1+s1s2). The difference between the ν-sign and ν-sign weights is a finite set of half-lines and points (L_O), plus the full lines U_H.
  - 15 of the 105 companions have α on an axis, where some ν-signs are 0. They are listed in data/slaw.json; see NOTES.
- **Odd pair (c1 = (−1,1), c′ = (−7,2)).**
  - Completion: ½(E(√10·ν1·√y) + E(√35·ν2·√y)).
  - Shadow: weight-3/2 unary thetas. One is Σ n e(5n²τ/2) (n ∈ α1+Z) paired with θ_{7/2}. The other is Σ m e(35m²τ/4) (m ∈ α2+Z) paired with θ_1(ν1+7ν2/2), glued over 2 classes.
- **Axis half-lines (Q = 1, 7/2).** These are positive-definite 1-D pieces. Vignéras' equation has no bounded weight-½ solution other than constants, so they have no completion on H; they are quantum modular (false/partial thetas). Their quantum shadows are Σ n e(n²τ) (n ∈ α1+7α2/2+Z) and Σ n e(7n²τ/2) (n ∈ α1+α2+Z). *Argument sketch.*
- **Even double orthant.** This piece is not of Zwegers form, for two reasons (*argument sketch*):
  - EVEN is even under ν → −ν, while Zwegers' ρ is odd.
  - The natural product candidate ½(1+E1E2) violates Vignéras' equation: its cross term is ∝ B(c1,c′) = −35 ≠ 0.

  The orthant is also not a union of unit-group fundamental cones, because Q(e1) = 1 ≠ Q(e2) = 7/2 (automorph [[−1,−7],[2,13]]). So EVEN is a partial Hecke-type sum.

Category B (numerical)

Interior S-law tests (120 τ, fixed 35-component law vector S-matrix, $\hat{H}(-1/\tau) = (\tau/\sqrt{35})\cdot M\cdot\hat{H}(\tau)$):

object	r	verdict
gating: $\hat{H}$ (cone part, completed)	6.0e−16	pass (required)
gating: holomorphic H_ν	0.300	fail (required)
O1: odd orthant pair $\hat{G}_O$	6.1e−16	**pass**
E1: even product candidate $\hat{E}$ = ½(1+E1E2)	0.903	**FAIL** (as predicted)
O3: completed orthant ¼(1+E1)(1+E2)	0.451	**FAIL** (as predicted)
F1: completed full series $\hat{S}_\nu = ½\hat{E} + ½\hat{G}_O − \hat{H}$	0.451	**FAIL** (as predicted)
F2: residual($\hat{S}$) = ½·residual($\hat{E}$)	max diff 4.7e−16	**pass**

Truncation error (box 40 vs 50) is ≤ 6e−17.

Finite-difference shadow tests (4 τ × 6 companions): - **O2:** the odd-pair shadow formula matches to 3.4e−9 (**pass**). - **E2:** the product-rule $\partial_{\bar\tau}$ of $\hat{E}$ matches to 3.3e−9 (**pass**; this is a calculus check). Its $\partial_{\bar\tau}$ carries E-factors, so it is not a sum of unary-theta products.

Radial test R1 (x ∈ {1, ½, ⅓, ⅔}, 6 companions, t_j = 2^{−j}·0.02/(2m)²): - For the completed $\hat{S}_\nu$, the max |c1| is 5.6e−10 and the max |c2| is 1.6e−8. Both are ≤ 1e−6, so **pass 4/4**: the completed A1 and A2 vanish. - Control: the uncompleted S_ν gives |c1| from 0.48 to 1.13 (> 1e−3, as required).

Verdicts

- Odd orthant completion: obeys the fixed S-law. **Pass.**
- Even double orthant: the proposed product completion **fails**. The even piece has no weight-1 Zwegers/Vignéras completion of this type.
- Completed full series: **fails** the fixed S-law. The whole failure equals ½ × the failure of the even piece (F2).
- A1/A2: after any smooth completion they vanish identically (R1), so radial “closure” is automatic and does not discriminate. The obstruction is visible only in the interior S-law.
- **Shadow rule (preregistered): the full-series shadow is NOT located.** What is located: the cone part plus the odd orthant, with four weight-3/2 unary thetas ($5n^2/2$, $R^2/34$, $35m^2/4$ glued to the walls 7/2, 595/2, 1). Not completed: the even double orthant and the axis half-lines.

Appendix: order6 notes (descriptive; disclosures)

1. **Axis companions.** 15 companions have α1 ∈ Z or α2 ∈ Z, so ν can lie on an axis, where sgn 0 = 0 gives weight ½. PREREG §1 required these to be listed (data/slaw.json, “axis_alphas”). For them, the ν-to-ν correction L_O has extra half-

lines on the axis. The completed functions and the S-law tests are unaffected, because they use ν-signs throughout.

2. **Dropped descriptive check.** The check in PREREG §3 that "y^{1/2}∂_τÊ is not antiholomorphic" was not run: its statement was ill-posed, since Zwegers shadows also carry a holomorphic θ factor. No verdict depended on it.
3. **slaw.py edit.** A garbled multiplier line in shadow_O was cleaned up before hashing and before any run. The value is unchanged: the factor is 1.
4. **Reading (Category B).**
   - Ê fails with r ≈ 0.90, and the completed orthant and full series fail with exactly half of that (F2). So the entire remaining S-law defect of the full series sits in the even double-orthant piece.
   - A modular completion of the even piece would need a different object: for example a non-Vignéras weight-1 completion, a weight-0 Maass/Hecke-type form attached to Q(√35), or an enlargement of the 35-component law vector space. This is open.
5. **Process.** nice, OMP_NUM_THREADS=1, ulimit -v 3000000. No push, no external sends. New files only.